4.1.1 Attraction and repulsion of charges

Like charges repel and unlike charges attract, with equal force magnitudes on the two objects. Attraction alone does not prove opposite net charges, since charge redistribution can make a charged object attract a neutral one. Repulsion provides direct evidence that both objects carry charges of the same sign.

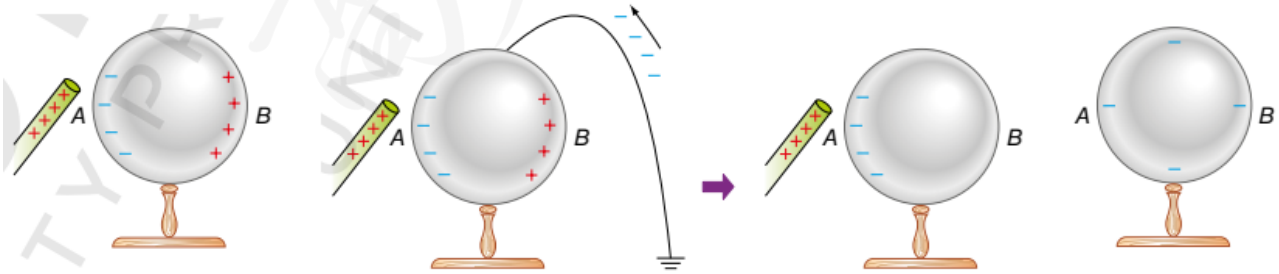
4.1.2 Conservation of charge

Net charge remains constant in an isolated system. Charging transfers or separates charge rather than creating it. Rubbing can leave two objects with equal and opposite charges. When conductors touch, total charge is conserved, not each conductor's original charge. The final division need not be equal and depends on their geometry and size.

4.1.3 Electrostatic induction and charging

A nearby charge slightly separates positive and negative charge in an insulator or redistributes free electrons in a conductor. The nearer opposite charge experiences stronger attraction, so the object can be attracted while remaining neutral overall.

Charging by induction uses earthing to change a conductor's net charge. Hold a positive rod near a neutral metal sphere and earth the sphere: electrons flow from Earth into it. Disconnect the earth first to retain the extra electrons, then remove the rod. The sphere is left negatively charged. With a negative rod, electrons flow to Earth and the same sequence leaves the sphere positively charged.



Left to right: a positive rod redistributes electrons; earthing with the rod held lets electrons enter; disconnect earth before removing the rod, leaving a negative sphere.

4.1.4 Coulomb's law

For point charges in vacuum, or approximately in air, force magnitude is proportional to the product of charge magnitudes and inversely proportional to separation squared. It acts along their joining line. Calculate the magnitude first, then determine attraction or repulsion from the signs. With multiple charges, add the individual forces as vectors.

$$F = \frac{|\, Q_1 Q_2 \,|}{4\pi\varepsilon_0 r^2}$$

F: force magnitude (N); Q_1, Q_2: charges (C); r: separation (m); $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$. Point charges in vacuum/air; like signs repel, opposite signs attract.

4.1.5 Definition of electric field

Electric field is electric force per unit positive test charge. Its direction is the force direction on a positive charge. A negative charge feels the opposite force. A test charge must be small enough not to appreciably disturb the source distribution. Using a larger test charge does not itself make the original field stronger.

$$\vec{E} = \frac{\vec{F}}{q_0}$$

E: electric field (N C⁻¹); F: force on a small positive test charge (N); q_0: test charge (C). A negative charge experiences force opposite to E.

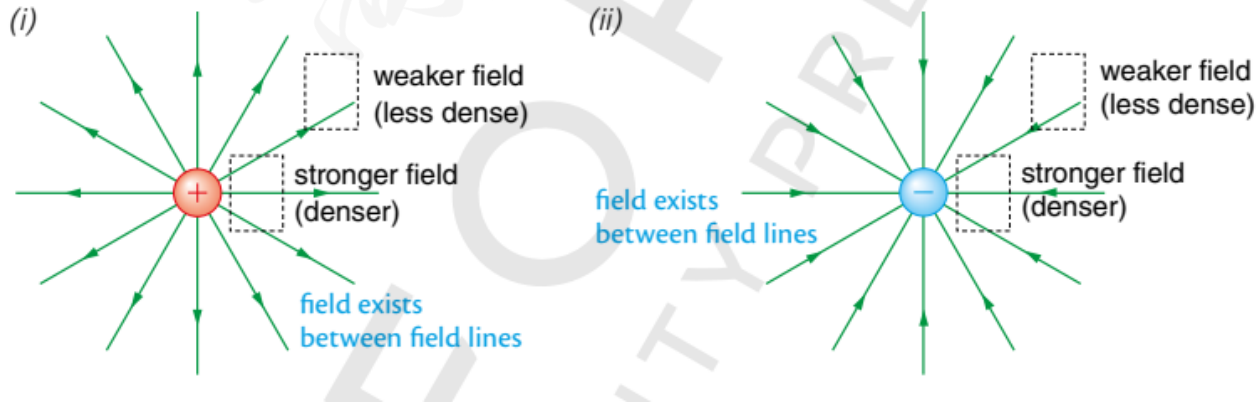
Electric field of a point charge

A point charge's electric field magnitude is proportional to source charge magnitude and inversely proportional to distance squared. It points outwards from positive charge and inwards towards negative charge. Add fields from multiple sources as vectors at the specified point. A zero resultant field does not imply absent charges or zero individual fields.

* Extension question: with potential zero at infinity, an isolated point charge has $V = \frac{Q}{4\pi\varepsilon_0 r}$, inversely proportional to distance. For the same charge, $V_Y/V_X = r_X/r_Y$. This is the asterisked application in 2014 Paper 1A Q23.

$$E = \frac{|\,Q\,|}{4\pi\varepsilon_0 r^2}$$

E: field magnitude (N C⁻¹); Q: source charge (C); r: distance (m); $\varepsilon_0 = 8.85 \times 10^{-12}$ C² N⁻¹ m⁻². Field points away from +Q and towards −Q.



Field lines point outwards from positive charges and inwards towards negative charges.

4.1.6 Uniform field between parallel plates

Between large parallel plates, away from edges, the field is approximately uniform. Its magnitude is potential difference divided by plate separation and its direction is from higher to lower potential. At fixed voltage, reducing separation strengthens the field. Multiply field by particle charge to find force. Negative charges are forced opposite the field direction.

$$E = \frac{V}{d}$$

Uniform field between large parallel plates, away from edges. E: magnitude (V m^{-1}); V: potential difference magnitude (V); d: separation (m). Field points from high to low potential.

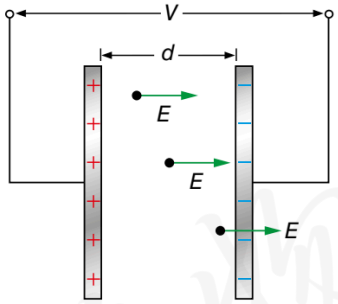
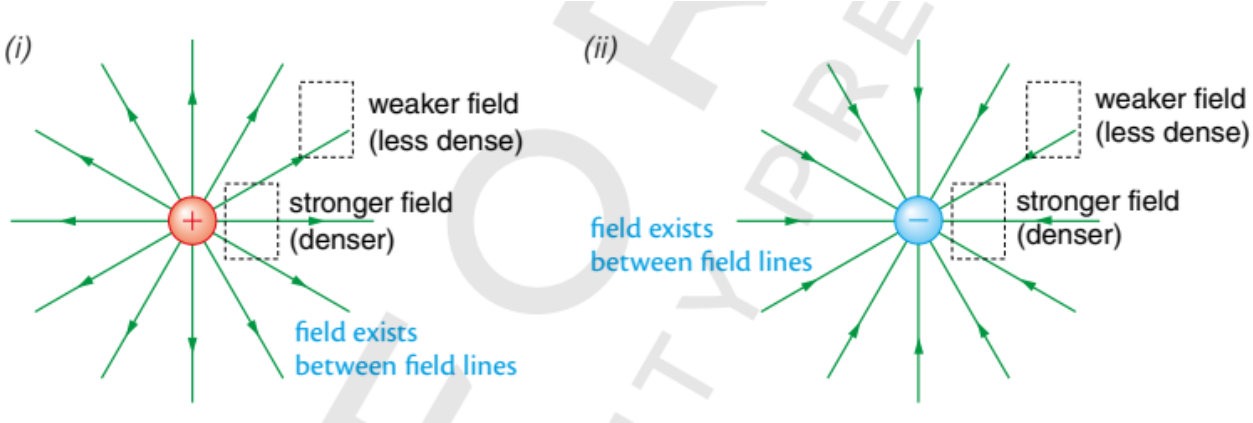


Fig 1.3j *E is the same everywhere between two parallel charged plates.*

The field is approximately uniform away from edges and points from the positive plate towards the negative plate. d is the perpendicular plate spacing.

4.1.7 Reading electric field lines

Electric field lines start from positive charges and end at negative charges. Their arrows show the field direction. Closer spacing indicates a stronger field, and field lines never intersect.



Field lines point outwards from positive charges and inwards towards negative charges.